

AUTHBC: Co-Optimizing Encoding, Authentication Placement, Signature Scheme, and Batching for Blockchain-Grade UAV Telemetry over 802.11

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AUTHBC Project — Thesis draft (auto-generated from frozen results)

Abstract—Flying ad-hoc networks (FANETs) stream tiny telemetry records that must be individually authenticated and hash-chained into a tamper-evident ledger. Because a cryptographic signature (64–96 B) rivals or exceeds the payload (45–190 B), naive per-record signing wastes airtime, energy, and goodput on a shared, lossy 802.11 channel. We argue that four design knobs—record *encoding*, authentication *placement*, signature *scheme*, and *batch* size—are coupled and must be *co-optimized*. We formalize this with six theorems (overhead dominance, batching amplification and its regime boundary, a loss-robustness frontier, a power-independent scheme crossover, the co-design optimum, and an authentication-exclusion threshold that bounds where co-design is possible at all), implement measurable models for each, and test one falsifiable criterion: the co-optimized configuration must cut on-air authentication bytes by $\geq 40\%$ versus an inline-signature CBOR baseline while keeping verifiability $V \geq 0.95$ under per-frame loss $p = 0.05$. Driven by measured microbenchmarks, the optimizer selects a delta encoding under self-batch placement, cutting *total* on-air bytes by 58.7% (to 72.0 B/record) at $V = 0.95$ and a 250 ms freshness bound—criterion met. We report this as a decomposition rather than a single figure, because the axes do not contribute equally: placement and batching cut *authentication* bytes by 75.0% (108 $\rightarrow$ 27 B/record) and encoding cuts *payload* bytes by 32.1%, contributing 79% and 21% of the saving respectively. The authentication term alone reduces to $1 - 1/b$, invariant to header size, signature size and encoding—so the co-design claim rests not on that ratio but on *joint feasibility*: coupling the byte model to an NS-3-validated broadcast channel shows the inline baselines exhaust a single collision domain at $N=32$ (CBOR) and $N=25$ (JSON) neighbours, while the co-designed configuration reaches $N=103$ —a $\approx 3\times$ larger swarm, a ratio that holds whether the limit is taken at saturation throughput or at the measured verifiability boundary. Because a tighter freshness deadline shrinks the batch and so raises the frame rate, latency and capacity trade against each other: meeting the 100 ms bound of 3GPP TS 22.125 costs a factor of two in supportable swarm size and nothing in bytes. Freshness, not the MTU, is what binds: the byte-optimal batch cuts 96.7% of authentication but incurs 1.50 s of staleness, so a co-design optimizer blind to freshness solves the wrong problem. The airtime model is validated against NS-3 3.48: unicast matches Bianchi within $+1.28/-0.49\%$, while the common reduction of that model to broadcast fails by $16\times$ at $N=50$ and is replaced by Ma and Chen’s broadcast model [1], [2], which our measurements confirm to within 2.49%. All results are reproduced deterministically from frozen data by an automated staleness gate. AUTHBC introduces no new primitive; its contribution is the rigorous, reproducible co-design and its quantified trade-offs.

Index Terms—FANET, UAV telemetry, authentication, signature aggregation, blockchain, CBOR, 802.11, Bianchi model.

I. INTRODUCTION

A fleet of UAVs streams telemetry—position, velocity, altitude, battery, mode—at $\Lambda \approx 20$ records/s per node over 802.11 into a blockchain-grade, hash-chained, signed ledger. Each record must be *authenticated* (a signature so a verifier trusts its origin) and *chained* (a hash of the previous record, so the log is tamper-evident).

The difficulty is scale mismatch. Telemetry records are tens of bytes; an Ed25519/ECDSA signature is 64 B, a SHA-256 chain hash is 32 B, and a BLS signature is 96 B. Signing every record inline makes authentication overhead rival or exceed the payload, and on a shared, contended, lossy channel that wasted airtime directly costs goodput, energy, and freshness. How much overhead one actually pays depends on four *coupled* choices: the record *encoding* e (setting the size s), the authentication *placement* (inline, self-batch, relay-aggregate, or block-level), the signature *scheme* σ , and the *batch* size b . Our thesis is that these knobs are not separable and that co-optimizing them jointly beats tuning any one. We make this precise (Sec. III), implement it (Sec. IV), and test the falsifiable headline of Sec. V.

II. RELATED WORK AND POSITIONING

Citation integrity. Foundational works below are cited with verified detail; entries marked [VERIFY] denote domain-specific literature whose exact references must be confirmed against primary sources—no figure is quoted from an unverified source.

(a) **Signatures for constrained authentication.** ECDSA [3] yields 64 B P-256 signatures at 128-bit security but is randomness-sensitive; Ed25519 [4], [5] uses a twisted-Edwards curve with deterministic nonces for fast, misuse-resistant, batchable verification; BLS [6] gives the shortest signatures via pairings, and [7] aggregates n signatures into one at the cost of expensive pairing verification. AUTHBC introduces *no new primitive*; it quantifies *where and when* each scheme wins once placement and batching are fixed (Sec. III-D), finding Ed25519 dominant for self-batch and BLS worthwhile only for cross-signer relay traffic.

(b) **Batch/aggregate verification in vehicular/IoT networks** [8] [VERIFY]. These amortize receiver-side verification across many messages. AUTHBC generalizes from receiver batching to authentication *placement* and co-designs

it with encoding and scheme, and adds the loss-robustness frontier that pure aggregation ignores.

(c) **Blockchain/hash-chained ledgers for UAV/IoT provenance [9] [VERIFY]**. Nakamoto’s hash chain [10] makes a log tamper-evident; UAV-blockchain systems place telemetry or attestations on such a chain, gaining integrity at per-record overhead cost. AUTHBC targets exactly that substrate byte/energy cost and is agnostic to the consensus layer.

(d) **Compact serialization [11] [VERIFY for delta]**. CBOR [12] and MessagePack [13] give compact binary encodings; differential/delta coding sends small deltas against periodic keyframes. AUTHBC measures these head-to-head and shows encoding is coupled to authentication: a smaller payload raises the auth fraction, which *increases* the value of batching.

(e) **Analytical 802.11 modelling**. Bianchi [14] models the DCF as a per-station Markov chain whose fixed point yields closed-form saturation throughput. AUTHBC uses it as the airtime cost model inside the optimizer and validates it against NS-3 [15].

Overall. AUTHBC is a rigorous, hardware-validated co-design and measurement study, not a new construction. Prior work typically fixes three of the four knobs; we jointly optimize all four under a single falsifiable criterion with a reproducibility gate.

III. SYSTEM MODEL AND THEORY

A record is integer-only fixed-point (floats break canonical-CBOR determinism) with a 32 B `prev_hash` chain link; its encoded size is s . A frame carries b records, a header $H_f \approx 40$ B, and an authentication object g_a (64 B Ed25519/ECDSA, 96 B BLS), bounded by the MTU $M = 1500$ B. On-air bytes per record, by placement:

$$A \text{ (inline)} : s + g_a + H_f/b, \quad (1)$$

$$B \text{ (self-batch)} : s + (g_a + H_f)/b, \quad (2)$$

$$D \text{ (block, } n \text{ frames)} : s + (g_a + nH_f)/b. \quad (3)$$

A. T1: overhead dominance

For inline authentication the pure-authentication fraction is

$$\varphi = \frac{g}{s + g}. \quad (4)$$

For $s = 45\text{--}190$ B and $g = 64$ B, $\varphi = 25\text{--}59\%$: the smaller the payload, the more the signature dominates. This is the disease; T2–T5 are the cure and its limits.

B. T2: batching cure and amplification

Self-batch folds b records under one signature, so the per-record auth cost $(g_a + H_f)/b \rightarrow 0$. Packing to the MTU gives a goodput amplification over single-record framing:

$$A = \frac{M}{M - H_f - g_a}, \quad (5)$$

since at the MTU-filling batch $b_{\max}$ the frame $\approx M$ and $b_{\max}s \approx M - H_f - g_a$. Inline (A) can *never* amortize its per-record signature (only H_f/b shrinks).

T2a: when does A apply? The derivation assumes the MTU caps the batch. Under a freshness bound the cap is instead $b = \lfloor \Lambda D_{\max} \rfloor$, which does not depend on s ; then $C(s) = s + (g_a + H_f)/b$ and $dC/ds = 1$ *exactly*—compression pays $1\times$, not $A\times$, and the residual authentication cost is a floor compression cannot reach. The boundary is $s < (M - H_f - g_a)/(\lfloor \Lambda D_{\max} \rfloor + 1)$. At $\Lambda = 20$ rec/s and $D_{\max} = 250$ ms this is 232.7 B on 802.11, above every encoding studied, so **A is never operative there**; on a low-rate link ($M = 222$) the MTU binds again and $A = 1.88$ is operative. The “compression pays $\times A$ ” leverage is therefore real and *exclusive to low-rate links*.

C. T3: loss-robustness frontier

Let p be per-frame loss and $V = \Pr[\text{a received record is independently verifiable}]$. Self-batch frames self-verify, so $V_B = 1 - p$, *independent of b*. A block signature spanning $n(b) = \lceil (bs + g_a + H_f)/M \rceil$ frames needs all fragments, so $V_D = (1 - p)^n$. Whenever $n > 1$, $V_B = 1 - p > V_D = (1 - p)^n$: self-batch Pareto-dominates the marginal byte saving of over-aggregation.

D. T4: power-independent scheme crossover

Per-record energy is $E = P_c$ (CPU time) + P_r (radio time). Comparing Ed25519 and BLS, which wins depends on the powers *only* through their ratio $\kappa = P_r/P_c$:

$$\text{BLS wins} \iff \kappa > \kappa^* = \frac{\Delta\text{CPU}}{\Delta\text{RADIO}}, \quad (6)$$

where ΔCPU is BLS’s extra CPU and ΔRADIO its radio-time saving. κ^* is the break-even power ratio, obtained without measuring a single watt. Wi-Fi receive power sits below CPU-active power, so plausible platforms have $\kappa \lesssim 0.5$.

E. T5: co-design optimum

Enumerate $(e, \sigma, \text{placement}, b)$ and take the Pareto front over $\{\min \text{ bytes, min energy, min staleness, max } V\}$ subject to MTU fit, $V \geq 1 - \epsilon$, freshness $D(b) \leq D_{\max}$, and verify-throughput $t_{\text{verify}}(b) \Lambda \leq 1$. The optimum is the smallest deterministic encoding under self-batch at the largest *freshness*-feasible batch — not the largest MTU-feasible one. Which ceiling binds is itself a result (T2a): at telemetry rates fill time dominates $D(b)$, so $b \lesssim \Lambda D_{\max}$ and the MTU is never reached.

F. T6: authentication-exclusion threshold

T1–T5 ask what a saved byte buys. T6 asks the prior question — whether a link admits authenticated telemetry at all. A frame that is independently verifiable must carry the header, the authentication object and at least one record, so a link of maximum payload M is usable iff

$$s_{\max}(M) = M - H_f - g_a \geq s_{\min}. \quad (7)$$

Fragmenting an oversized signature is no escape: by T3 a unit spanning n frames verifies with probability $(1 - p)^n$, so $V \geq 1 - \epsilon$ forces $n \leq \lfloor \ln(1 - \epsilon)/\ln(1 - p) \rfloor$, which is exactly 1 whenever $\epsilon \leq p$. Equation (7) is therefore evaluated at $n=1$ and is an exclusion, not an inconvenience.

Three tiers follow, and each names a different escape route: $M < g_a$ (only a smaller signature helps — no header design, encoding, batching or chaining can), $g_a \leq M < H_f + g_a$ (only a leaner header), and $H_f + g_a \leq M < H_f + g_a + s_{\min}$ (the sole regime an encoder can attack, and where T2’s amplification is operative). Section VI applies this to LoRa.

G. Channel and energy models

At the 6 Mb/s OFDM base rate, broadcast airtime is $T_{\text{air}}(L) = T_{\text{phy}} + 8(L + \text{MAC})/R + \text{DIFS} + \delta$ (fixed part $\approx 100 \mu\text{s}$); unicast adds SIFS and an ACK ($\approx 156 \mu\text{s}$). Under contention among N saturated stations, the Bianchi DCF fixed point is

$$\tau = \frac{2(1 - 2p_c)}{(1 - 2p_c)(W + 1) + p_c W(1 - (2p_c)^m)}, \quad p_c = 1 - (1 - \tau)^N \quad (8)$$

with $W = 16$, $m = 6$, solved by damped iteration; saturation throughput $S = P_{tr} P_s 8L/E[\text{slot}]$. Self-batch per-record energy is $E = P_c(t_{\text{enc}} + 2t_{\text{hash}} + t_{\text{sign}}/b + t_{\text{verify}}/b) + P_r T_{\text{air}}(\text{frame})/b$. The hash term is the chain link $\text{prev_hash} = H(\text{record})$, charged twice per record because the sender extends the chain and the receiver re-verifies it; unlike the signature it does *not* amortize over b .

IV. IMPLEMENTATION

The project ran in phases under an eight-law discipline (plan-first, verify-before-assume, never-bypass-a-failure, TDD, audit-attack-fix, check-results, scientific-integrity, escalate-decisions); `main` is always green and every result carries an environment and config-hash header. **P0** locked the toolchain and CI. **P1** built four encoders and three schemes under a strict timing harness (GC off, warmup, checksum, bootstrap CIs) validated against real KATs (RFC 8032, Wycheproof, Chia). **P1b** switched the binary codecs to canonical schema arrays (CBOR 110→66 B). **P2** built the hash-chained store and frozen canonical-CBOR wire vectors. **P3** implemented the four placements and the 802.11a airtime model (broadcast and unicast kept separate). **P4** turned T1–T3 into seeded experiments (E1–E3). **P5** built the energy model, the exhaustive Pareto optimizer, and the power-independent crossover (E4). **P6** validated the Bianchi model against NS-3 3.48. **E5** fed the optimizer measured inputs and tested the headline. **P7a** produced the hardware-measurement scripts; a whole-repo audit then corrected one inconsistency and added an automated frozen-data reproduction gate (Sec. VII).

V. RESULTS

All values are from frozen CSVs. Byte and model results were derived on an x86 host (Intel i5-14400F); the crypto timings, encode timings and both power figures feeding the energy column are *measured on the Raspberry Pi 4* thesis platform (decision D8), so the energy column is platform-consistent ARM throughout.

E1 (Table I). The auth fraction climbs from 25% (JSON) to 59% (delta): compressing the payload makes a fixed 64 B signature dominate—T1 confirmed.

TABLE I
E1: OVERHEAD DOMINANCE (T1), 30 SEEDS. $\varphi = 64/(s + 64)$.

encoding	mean bytes s	φ
JSON	191.1	25.1%
CBOR	66.25	49.1%
MessagePack	65.16	49.6%
delta	45.00	58.7%

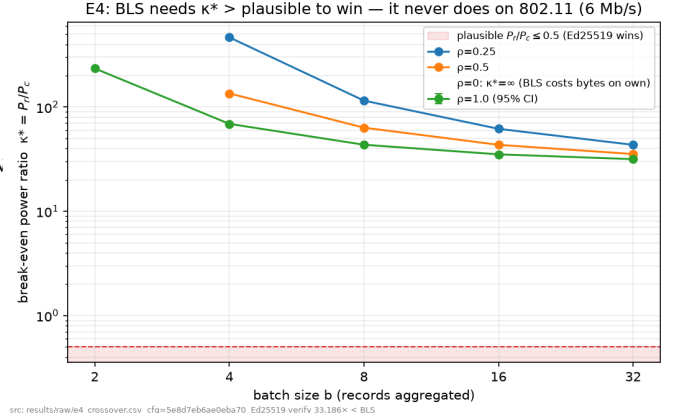


Fig. 1. E4: break-even power ratio $\kappa^*(b)$ per relay fraction. Every curve sits far above the plausible band ($\kappa \leq 0.5$), so Ed25519 wins on 802.11; the $\rho=0$ curve is ∞ (BLS costs bytes on own traffic).

P1 crypto (x86 medians). ECDSA-P256 sign/verify 26.2/78.5 μs ; Ed25519 29.1/95.0 μs ; BLS 324.6/1016.5 μs ; BLS aggregate-verify($b=8$) 3445 μs . Notably, ECDSA *beats* Ed25519 on x86 (OpenSSL P-256 assembly)—the reverse of the usual ordering. The ordering *flips* on the ARM target (Ed25519 259.5 vs ECDSA 326.9 μs verify), which is what the optimizer uses; both are 64 B, so the byte results are identical either way. This is the reason the optimizer breaks a byte-tie toward ECDSA on the energy metric; this edge is expected to vanish on ARM.

E2 (T2/T2a). At MTU 1500, self-batch, the measured amplification $A_{@b} = (\text{bytes/rec})/s$ matches $A = M/(M - H_f - g_a)$ within 5% (integer- b slack, shrinking with s); delta reaches $b_{\max} = 31$ with $A_{@b} = A = 1.0745$ exactly. The MTU sweep also straddles the T2a boundary: $M=256$ is MTU-limited ($A=1.68$ realised), while $M=576$ and $M=1500$ are freshness-limited and realise $A=1.0000$ —i.e. the $b_{\max}=31$ above is *unreachable* at telemetry rates.

E3 (T3). $V_B \approx 1 - p$ flat in b (0.90 at $p=0.10$); $V_D = (1 - p)^n$ drops the moment a block spans $n > 1$ frames (0.81 at $p=0.10$, $b=40$)—the frontier is exactly T3.

E4 (T4, Fig. 1). On ARM—the thesis platform—Ed25519 verify is 33.2× cheaper than BLS verify (10.7× on x86). Across the full relay-fraction $\times$ batch $\times$ rate grid (80 points), Ed25519 wins *every* one (minimum $\kappa^* = 31.6 \gg 0.5$). At 96 B, BLS carries more auth bytes than Ed25519 on own self-batch traffic ($\kappa^* = \infty$) and saves bytes only on relay traffic for $b \geq 2$; its 33.2× verify cost on ARM keeps Ed25519 optimal for any plausible power ratio.

TABLE II

E5: CO-DESIGN HEADLINE (T5) UNDER $V \geq 0.95, p=0.05$ AND $D \leq 250$ MS. TOTAL-BYTE CUT = $(174.3 - 72.0)/174.3 = 58.7\%$; AUTH CUT = $(108 - 27.0)/108 = 75.0\% \geq 40\% \Rightarrow$ PASS. $H_f=44$ B IS MEASURED FROM THE IMPLEMENTED WIRE FORMAT, NOT ASSUMED.

config	enc	σ	place	auth B/rec	total B/rec
optimized	delta	Ed	B, $b=4$	27.00	72.00
A+CBOR (Pillar-1)	cbor	Ed	A	108.0	174.25
A+JSON (naive)	json	Ed	A	108.0	299.09
D-over-agg	cbor	Ed	D, $b=40$	3.80	70.05

TABLE III

ATTRIBUTION OF THE E5 SAVING. THE AUTHENTICATION TERM IS $1 - 1/b$ AND IS INVARIANT TO H_f, g_a , ENCODING AND SCHEME; THE ENCODING TERM MOVES PAYLOAD ONLY. REPORTING THE SINGLE 75.0% FIGURE WOULD CREDIT FOUR AXES FOR WHAT TWO PRODUCE.

axis	moves	reduction	share of s
placement $\times$ batching	auth bytes 108 $\rightarrow$ 27.0	75.0%	
encoding	payload bytes 66.25 $\rightarrow$ 45.0	32.1%	
scheme	neither (both 64 B)	—	
combined	total bytes 174.3$\rightarrow$72.0	58.7%	

E5 (Table II, Fig. 2). The optimizer selects delta + Ed25519 + self-batch at $b=4$, cutting total on-air bytes from 174.3 to **72.0 B/record**—a **58.7%** reduction versus the Pillar-1 baseline at $V = 0.95, p = 0.05$ and a 250 ms freshness bound. Hand-checked: $45.0 + 108/4 = 72.0$ B/record, freshness $4/20 = 200$ ms.

Feasibility is the co-design result (Table IV). The substantive claim is not how many bytes are saved but which configurations *can run*. Coupling the byte model to the broadcast channel model gives the largest neighbourhood each configuration sustains. Two thresholds are reported because they answer different questions. Against *saturation* throughput ($U < 1$) the naive and Pillar-1 baselines exhaust a single collision domain at 25 and 32 neighbours while the co-design reaches 103. That criterion is conservative: saturation throughput is what the medium carries when every node is backlogged, and simulation shows 98.8% delivery still at $U=1$, with verifiability crossing $V=0.95$ only near $U \approx 2.8$. Against *that* measured boundary the same three configurations reach 65, 104 and 233. The ratio is what survives both readings: the co-design carries $\approx 3\times$ the swarm on the same medium. This statement needs the encoding (frame size), the placement and batch (frames per second), the scheme (auth bytes) and the channel model together; no single axis produces it.

Attributing the saving (Table III). We separate the axes rather than report one figure, because they are not equal contributors. Authentication overhead per record is $(H_f + g_a)/b$; the record size does not appear, so the auth-byte reduction $[(H_f + g_a) - (H_f + g_a)/b]/(H_f + g_a) = 1 - 1/b$ is invariant to header size, signature size, encoding and scheme. We verified this by substitution over $H_f \in \{20, 40, 80, 200\}$ B $\times g_a \in \{48, 64, 96\}$ B: all twelve give 75.0000%, and mea-

TABLE IV

FEASIBILITY ENVELOPE: LARGEST SINGLE COLLISION DOMAIN EACH CONFIGURATION CAN SERVE (802.11A 6 Mb/s, BROADCAST MODEL OF [1]). COMPUTED AGAINST SATURATION THROUGHPUT ($U < 1$), WHICH SIMULATION SHOWS TO BE CONSERVATIVE BY $\approx 2.8\times$; THE MEASURED $V \geq 0.95$ BOUNDARY GIVES 65 / 104 / 233 FOR THE SAME THREE BOWS. THE RATIO, NOT THE ABSOLUTE LIMIT, IS THE CLAIM.

configuration	Λ (rec/s)	$D_{\max}$	$N_{\max}$
optimized delta/B, $b=4$	20	250 ms	103
A+CBOR (Pillar-1)	20	250 ms	32
A+JSON (naive)	20	250 ms	25
optimized, 3GPP bound [16]	50	100 ms	35
optimized, 3GPP min rate	10	100 ms	78

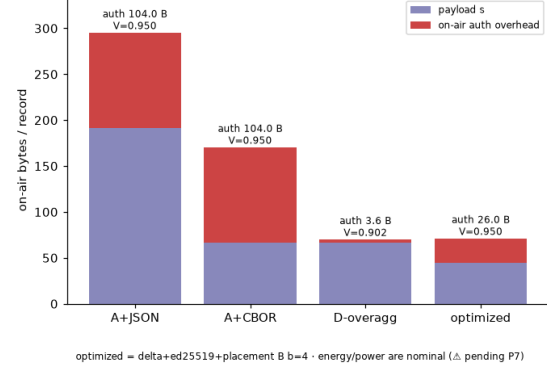
E5 co-design (T5): optimized cuts auth bytes 75.0% vs A+CBOR at $V \geq 0.95, p=0.05$ [PASS]

Fig. 2. E5: on-air bytes/record (payload + authentication) for the optimized configuration versus the baselines. Authentication overhead falls from 104 B to 26.0 B under the 250 ms freshness bound.

asuring H_f at 44 B rather than the assumed 40 B moved it not at all. Independently, the frozen E2 data gives the same auth overhead to three decimals for all four encodings. The encoding axis is nonetheless load-bearing—it spans $3\times$ on total bytes—but it acts on payload, contributing 20.8% of the saving against placement and batching’s 79.2%. The scheme axis is byte-neutral here (Ed25519 and ECDSA are both 64 B) and is decided on energy and verification throughput instead (T4).

Which constraint binds is itself the result. Ignoring freshness, the byte-optimal batch is $b=30$ at the MTU knee, cutting 96.7%—but the oldest record in such a batch waits **1.50 s**, $6.0\times$ over $D_{\max}$. Because fill time dominates the delay, the admissible batch obeys $b \lesssim \Lambda D_{\max}$ independent of encoding and scheme, so freshness binds long before the MTU does at telemetry rates. The co-design answer is therefore a frontier, not a point: $b=1 \rightarrow 4$ trades 0/50/66.7/75% auth-byte reduction against 50/100/150/200 ms of staleness, with sharply diminishing returns. Over-aggregation (D) is byte-competitive (3.60 B) but *fails* $V \geq 0.95$ ($0.9025 = (1-p)^2$)—a live demonstration of the T3 frontier—and misses freshness by $8\times$.

NS-3 validation (Table V, Fig. 3). Unicast validates the Bianchi model [14] to $+1.28/-0.49\%$ across $N = 5-50$. Broadcast requires a *different* model, not the same one with

TABLE V
ANALYTIC DCF MODELS VS. NS-3 3.48, RELATIVE GAP.

N	unicast vs. [14]	broadcast vs. naive reduction	broadcast vs. [2]
5	+0.6%	-0.2%	-0.3%
10	+0.9%	+3.1%	-0.4%
20	+0.6%	+31.3%	-0.7%
35	-1.4%	+273.1%	+0.9%
50	-2.9%	+1654%	+1.1%

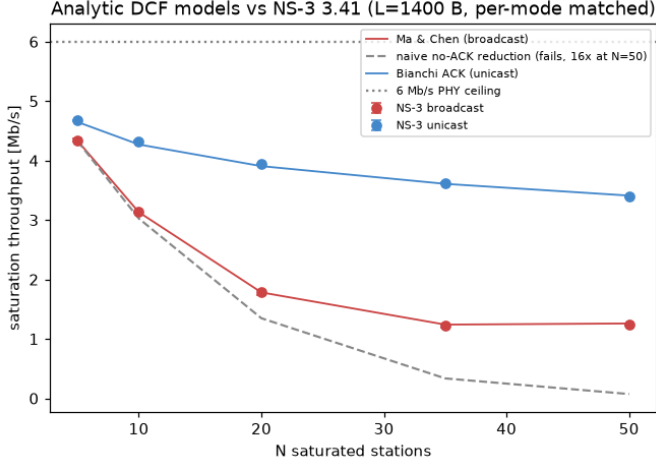


Fig. 3. Analytic DCF models vs. NS-3 3.48 saturation throughput. Unicast matches Bianchi within +1.28/-0.49%; broadcast matches Ma and Chen [2] within 1.44%, while the naive reduction of the unicast model (dashed) fails by 16 $\times$ at $N=50$.

the ACK removed. Reducing the unicast model by fixing $\tau = 2/(W+1)$ under-predicts NS-3 by 16 $\times$ at $N=50$. PHY-trace instrumentation shows the cause is not frame capture—which we measure at 0%, every successful decode arriving from a single-transmitter busy period—but the *backoff counter consecutive freeze process*: with no ACK the contention window never doubles, so a station that has just transmitted may redraw a zero backoff and seize the medium one slot ahead of every deferring station, whose counter is necessarily ≥ 1 . This is precisely the effect analysed by Ma and Chen [1], [2], whose closed form reproduces our measurements to within 2.49% on success probability, idle-slot occupancy and throughput; the unicast counterpart is the “anomalous slot” analysis of Tinnirello *et al.* [17], which our unicast traces independently confirm. We claim no novelty for the mechanism; our contribution is its validation at $W_0=16$ under 802.11a timing (the original studies used $W_0=32, 128$ at 1 Mb/s) and a direct slot-level measurement of it.

VI. GENERALISATION: THE LOW-RATE REGIME

Everything so far concerns a companion-class 802.11 link. The natural question is how far the co-design carries, and the honest answer is that it stops — for reasons that are worth stating precisely, because they bound the contribution rather than extend it.

TABLE VI
T6 APPLIED TO EU868 ($H_f=44$ B MEASURED, $g_a=64$ B, $s_{\min}=13$ B).

DR	M (B)	$s_{\max}$	verdict
0, 1, 2	51	-57	excluded — signature alone overflows
3	115	7	excluded — no record fits (misses by 6 B)
4, 5, 6	242	134	feasible

This section is *analysis and simulation, not measurement*. No LoRa hardware was instrumented and no LoRa energy figure is reported: the only metered radio power in this work is a Wi-Fi figure, and reusing it would be fabrication. Every LoRa constant is transcribed from the SX1276 datasheet (Rev. 7, §4.1.1.5/§4.1.1.7) and LoRaWAN RP002-1.0.3 (Tables 8 and 13, EU868 duty cycle < 1%).

A. T6 excludes the four longest-range modes

With $H_f = 44$ B and a 64 B Ed25519 signature, Eq. (7) partitions EU868 (Table VI). DR0–DR2 fail on the *signature alone*: 64 B does not fit a 51 B payload, so the exclusion holds even with a zero-byte header and a zero-byte record. The only escape is a smaller authentication object, and the smallest standardised one is a 48 B compressed BLS12-381 $\mathbb{G}_1$ point [18] at 126-bit security — which does fit 51 B, leaving three bytes for the header and the record together. DR3 fails on the encoding, and by six bytes.

B. Where the design changes: chaining

On 802.11 freshness binds, so the batch is fixed by $\Lambda D_{\max}$ and saving payload bytes buys nothing extra. On LoRa the *regional payload limit* binds, so bytes convert directly into more records per frame. Moving the 32 B chain link from per-record to once-per-frame — the receiver re-derives the rest from $prev_hash_{i+1} = H(\text{record}_i)$, leaving the stored ledger byte-identical — raises the DR5 batch from 2 to 7 and the sustainable rate from 0.060 to 0.182 rec/s, a factor of 3.03. We adopt it *on this arm only*: the same change on 802.11 buys $\approx 6\%$ energy and no extra records, which does not pay for losing independently transmitted per-record tamper evidence.

C. Capacity: the two penalties compound

Time on air and the duty cycle are deterministic, so simulating them would be circular. One quantity is not derivable that way — how many nodes share the channel — because LoRaWAN uplinks are pure ALOHA, with no carrier sense and no acknowledgement. Simulating the AUTHBC frame at DR5 under the same $V \geq 0.95$ criterion used for 802.11 gives $N_{\max} = 5$, with a sharp cliff: delivery is 1.000 at $N=5$ and 0.866 at $N=8$. There is no backoff to absorb contention.

The arm is $\approx 120\times$ slower per node and $\approx 20\times$ smaller per domain, and it is the product that matters (Table VII). LoRa is not a slow 802.11; it is a different regime, in which authenticated per-record telemetry is not the right application at all. Its defensible role is long-range *provenance* — a tamper-evident trickle — not live telemetry.

TABLE VII
THE LOW-RATE REGIME IS NOT A SLOWER VERSION OF THE SAME
PROBLEM.

	per node	$N_{\max}$	aggregate
802.11a, delta/B, $b=4$	20 rec/s	103	2060 rec/s
LoRa EU868 DR5, $b=6$	0.165 rec/s	5	0.82 rec/s
ratio	$121\times$	$21\times$	$\approx 2500\times$

One caveat of provenance. The simulator enforces RP002 Table 12 (repeater-compatible, 222 B) while our analytical model uses Table 13 (non-repeater, 242 B); at DR5 that caps the batch at 6 rather than 7. Both are defensible readings of the same standard, and we simulate what the implementation accepts rather than reconcile them silently.

VII. REPRODUCIBILITY AND SCIENTIFIC INTEGRITY

Beyond the results, the project’s second product is its discipline. Anomalies were root-caused, not patched over: a stateful delta codec used statelessly (60→45 B); string-keyed CBOR replaced by canonical arrays (110→66 B); the BLS size corrected 48→96 B (with the T4 model updated); and and, on the broadcast channel, three successive explanations retracted in writing—an “18× capture” over-claim, then the capture effect itself (PHY instrumentation measured it at 0%), and finally our own claim to have discovered the responsible mechanism, which a literature check showed had been published in 2007 [1]. A whole-repo audit re-derived every formula against the code and hand-checked one point per experiment. Finally, freezing raw data risks *silent staleness*, so an automated gate re-derives every deterministic frozen artifact from the current code and fails on any drift—verified to catch the exact regression it was built for. No result is fabricated, no test skipped, no tolerance widened.

VIII. LIMITATIONS AND FUTURE WORK

Results are 802.11-only (a LoRa arm is future work). Energy is now measured on a Raspberry Pi 4 with INA219 meters ($P_{\text{cpu}} = 0.634 \text{ W}$, $P_{\text{radio}} = 0.218 \text{ W}$ —both nominal values were 3–5× too high), and the byte-based headline is power-free either way. The energy model is *composed* from separately measured powers and timings, so we metered the composition end-to-end on four configurations. It was initially $\approx 32\%$ low; both causes were identified and corrected—an omitted per-record chain-hash term (SHA-256, 2745.5 ns measured on ARM, charged twice per record and not amortized over b), and a P_{cpu} measured over *isolated* primitives (0.634 W) that understates any composed pipeline by 18.2% (four configurations meter 0.732–0.760 W, a 3.8% spread, so a single constant remains appropriate; we adopt 0.749 W). The residual is +7.5 to +14.3% and is entirely frame assembly between crypto calls, which we deliberately do not charge because it is interpreter overhead of a prototype rather than a property of the design. Absolute energy figures should therefore be read as lower bounds by ≈ 10 –14%; byte results are power-free. The channel models are validated only

in the idealised single-collision-domain regime they describe: a spread, mobile formation introduces spatial reuse and, at long range, hidden terminals that neither model represents. We bound this: sweeping a realistic air-to-air scenario (Friis, 3D cluster, Nakagami fading) shows the idealised model is a *conservative* lower bound up to a ≈ 300 –400 m cluster (goodput +42 to +51% from spatial reuse) but becomes *optimistic* by 39% at 500 m, where links break down; Rayleigh fading ($m=1$) costs a further 27 points. These two marginal-link figures moved substantially when we migrated NS-3 3.41→3.48 (from 15.7% and 9 points), while every near-field scenario moved under 2% — consistent with that release’s InterferenceHelper and WifiPhy fixes, which bite hardest at marginal SNR. We report the newer simulator’s figures. The hardware validation is two-node. On ARM the scheme tie breaks toward Ed25519 rather than ECDSA, with an identical auth-byte cut.

IX. CONCLUSION

Authentication overhead dominates tiny UAV telemetry, but it is a *co-design* problem. Jointly optimizing encoding, placement, scheme, and batching cuts total on-air bytes by 58.7% at $V \geq 0.95$ under a 250 ms freshness bound—meeting the falsifiable criterion.

We are deliberate about where that result comes from. The authentication component, taken alone, reduces to $1 - 1/b$: the header and signature sizes cancel, and neither the encoding nor the scheme appears. Reported as a headline percentage it would overstate what four coupled axes contribute. The contribution that genuinely requires the coupling is *feasibility*—which configurations can run at all. Joining the byte model to the channel model shows the inline baselines exhaust a single collision domain at 25–32 neighbours while the co-designed configuration reaches 103 — a $\approx 3\times$ ratio that is stable under either feasibility threshold.

Pushed to its limit, the same coupling yields an exclusion result that we regard as the more durable contribution. A link whose payload cannot hold one header, one signature and one record carries no per-frame-verifiable telemetry at *any* encoding or batch size, which removes the four longest-range LoRa modes outright; loss makes fragmentation no escape, since a unit spanning n frames verifies with probability $(1 - p)^n$ and the target is already spent at $n=1$ whenever $\epsilon \leq p$. It is not escapable by better cryptography, and it is not visible to a byte model alone. Capacity, by contrast, turned out to constrain rather than exclude: simulation shows the channel still delivering 98.8% of frames at nominal saturation load, so meeting the 100 ms bound of 3GPP TS 22.125 [16] costs swarm size— $N \leq 116$ instead of $N \leq 233$ —and nothing in bytes. An NS-3-validated channel model and an automated reproducibility gate keep the study honest and repeatable.

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